MIL-P-70505 (AR)
AMENDMENT 2
20 October 1987
SUPERSEDING
AMENDMENT 1
20 January 1987

MILITARY SPECIFICATION

PROJECTILE, 155MM: XM864
PROPELLANT GRAIN

This Amendment forms a part of Military Specification MIL-P-70505 (AR), dated 28 December 1984, and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Add the following Military Standard:

MIL-STD-453 Inspection, Radiographic.

PAGE 3

3.3 Delete "930 ± 35 cal/g" and substitute "950 ± 50 cal/g".

3.4 Delete in its entirety and substitute the following:

"3.4 Strand burning rate. Strand burning rate tests shall be preformed as shown in TABLE I. Only results from the 70°F, 14.7 psia test shall be used for lot acceptance.

TABLE I. Strand burning rate.

Temperature (± 5°F)	Pressure (psia ± 10%)	Rate (mm/sec)
-50	14.7	INFO
70	14.7	1.00 ± .10
145	14.7	INFO

3.6.3 Delete in its entirety and substitute the following:

"3.6.3 Internal Defects The propellant grain shall contain no cracks, voids or fissures measuring more than .20 inch in any direction and no foreign material measuring more than 0.20 inch in any direction. The total combined length of all cracks, voids and fissures for the grain shall not exceed .50 inch and the combined length of all foreign material (maximum dimension each particle) shall not exceed 0.40 inch."

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3.7 Delete paragraphs a and b and substitute the following:

"a. Modulus of elasticity of 700 psi minumum per sample average."

b. Ultimate stress of 170 psi minimum per sample average."

PAGE 6

4.3.1 Change "TABLE I" to TABLE II".

4.3.1 Table I, change strand burning rate samples from "12" to "18". Add the following in the paragraph reference section of the strand burning test

"(14.7 psia tests only, 6 at each temperature)".

PAGE 10

4.4.2.3 Change title to "precast Inhibitor, Prior to assembly to grain". Delete special defect A and substitute the following:

"A. Inhibitor prior to bonding to propellant (applies to pre-cast inhibitor only) 100%, 3.6.2, Visual".

PAGE 11

4.4.2.4 Major 104: Change "four" to "two".

Major 106: Change "4" to "2".

Major 107: Change "8" to "4".

Minor 201: Change "Location of spacers" to "True position of spacers"

PAGE 12

4.4.2.5 Change Minor 202 from "100%" to ".65%".

PAGE 13

4.5.1 Delete in its entirety and substitute the following:

"4.5.1 Strand burning characteristics of propellant Along with propellant grains cast from each batch or mix of chemicals, nine test strands shall be taken from the same propellant batch and cured in the same curing chanber under the same conditions for the same length of time. The

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strands shall be tested in accordance with method T803.1 of MIL-STD-286 or an approved alternate (see 6.5). Three strands shal be tested at each of the temperature - pressure combinations in 3.4. If the average of three strands fails to meet the requirements of 3.4, the batch shall be rejected."

PAGE 15

4.5.5 Delete "3.6.2" in the last sentence and substitute "3.6". Add the following sentence at the end of the paragraph:

"Equipment and procedures used for radiographic inspection shall be in accordance with MIL-STD-453."

PAGE 16

4.5.10.3 Changed paragraph number to "4.5.6.3".

4.5.10.4 Change paragraph number to "4.5.6.4".

PAGE 17

6.4 Change "SMCAR-LCU-SI" to "SMCAR-FSA-IM".

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes, (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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